

Real-Time Taxi Demand Forecasting with Fare and ETA Prediction using Hadoop and Spark

First Review

Group-6

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23AID302 – Big Data Analytics

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Introduction to the Problem Domain

- Our project is in the area of Big Data Analytics applied to Smart Transportation — specifically taxi demand forecasting, fare estimation, and trip duration (ETA) prediction.
- Modern cities and ride-hailing platforms create massive data every second: pickup points, dropoff points, timestamps, trip distances, and fares.
- **Why this matters:**
 - Mismatch between taxi supply and passenger demand causes long passenger wait times and wasted fuel for idle drivers.
 - Unexpected weather changes and traffic spikes create price confusion and delays.
- Older, single-machine tools (Excel, Pandas, standard SQL) crash when handling 100+ million trip records or streaming live data.
- Big Data tools like Hadoop HDFS and Apache Spark are built to process this volume of data quickly and support real-time mobility decisions.

Problem Statement

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Current taxi demand forecasting, fare prediction, and ETA estimation tools are usually built as separate small-scale projects using static datasets. Most do not combine historical trip patterns with weather conditions, and cannot process live streaming data efficiently. Our project builds one unified Big Data framework using Hadoop HDFS and Apache Spark that cleans 12+ million records, handles schema drift, forecasts zone-wise taxi demand for the next hour, and predicts trip fare and ETA in real time.

Objectives

- Develop a scalable Hadoop–Spark framework for real-time urban transportation analytics using NYC TLC taxi and Open-Meteo weather datasets.
- Implement automated data cleaning at scale using Scala Spark to filter noisy records (negative fares, zero distance, GPS glitches) across 13 veracity rules.
- Enrich datasets with spatial and weather features using broadcast joins with 265 NYC Taxi Zones and hourly temperature/rain data.
- Train Spark MLlib machine learning models to predict hourly zone demand, trip fare, and trip duration (ETA).
- Build a real-time streaming pipeline using Apache Kafka and Spark Structured Streaming for live trip processing.
- Create an interactive decision dashboard with live demand heatmaps and fare/ETA prediction widgets.

Literature Review

Paper (Post-2025)	Method Used	Limitation / Gap
Zhang et al., <i>IEEE T-ITS</i> (2025)	Spatio-Temporal Graph Neural Network (ST-GCN) for dynamic taxi demand	Focuses only on demand; single-GPU model; ignores fare and ETA; no streaming.
Liu & Wang, <i>Appl. Soft Comput.</i> (2025)	Multi-task deep GRU with cross-attention for joint demand and fare	No ETA prediction; evaluated on static 1-month subset; no distributed HDFS ingestion.
Kumar et al., <i>ACM TKDD</i> (2026)	Weather-Aware Transformer for demand and travel time estimation	Excludes fare modeling; single-server setup; no real-time streaming alert dashboard.

Research Gap Identified:

- Existing post-2025 works focus on individual transport problems on small static subsets.
- They lack an integrated Hadoop–Spark framework, weather fusion, and unified triple prediction (Demand + Fare + ETA).

5Vs of Big Data

- **Volume:** Processes large-scale NYC Taxi datasets (17.25 GB raw, 12.1+ million trip records across 2022–2024) using Hadoop HDFS.
- **Velocity:** Spark Structured Streaming and Kafka enable near real-time processing of continuous trip streams.
- **Variety:** Integrates structured taxi trip Parquet files, spatial 265 Taxi Zone CSVs, and semi-structured Open-Meteo weather JSON.
- **Veracity:** Improves data reliability through 13 Scala Spark veracity cleaning rules, filtering 737,000+ noisy records.
- **Value:** Generates actionable mobility insights (Demand Heatmaps, Fare Quotes, ETA Alerts), enabling faster driver allocation and passenger planning.

Proposed System (1/3)

Overview

- The proposed framework integrates Taxi Demand Forecasting, Fare Prediction, and ETA Estimation into a single unified Hadoop–Spark architecture.
- Store multi-year NYC Taxi trip datasets and weather records in Hadoop HDFS.
- Perform data preprocessing including schema unification, cleaning, and feature extraction using Scala Spark.
- Enrich trip records with 265 Taxi Zones using broadcast joins to avoid network shuffle.
- Process streaming trip updates using Spark Structured Streaming and Kafka.

Proposed System (2/3)

Novel Framework Components

1. Unified Multi-Task Feature Engineering

- Derive temporal features (pickup hour, day of week, weekend/night flags).
- Derive spatial features (airport trips, zone demand counters).
- Derive rate features (average speed, fare per mile).

2. Weather & Calendar Context Fusion

- Join hourly temperature, rain (mm), and wind speed from Open-Meteo JSON.
- Label extreme rain events to predict weather-triggered demand spikes.

3. Spark MLlib Model Pipeline

- Train Random Forest and Gradient Boosted Trees for joint demand, fare, and ETA prediction.

Proposed System (3/3)

Mobility Intelligence Framework

1. Forecast Zone-Wise Hourly Taxi Demand.
2. Predict Individual Trip Fares.
3. Estimate Trip Duration and ETA.
4. **Real-Time Streaming Alert Engine**
 - Compare live Kafka stream pace with baseline forecasts.
 - Generate surge alerts during weather events or transit outages.

Final Outputs

- Live Demand Heatmap for Drivers.
- Fare & ETA Widget for Passengers.
- System Performance Benchmarks for Evaluators.

Implementation Details

Hadoop Cluster

- 1 NameNode + 3 DataNodes
- Hadoop HDFS 3.2.1
- Replication factor: 3
- Docker Desktop + WSL2

Processing Engine

- Apache Spark 3.5.3
- Spark Standalone / YARN
- Scala 2.12.18 engine code
- sbt 1.9.8 assembly JAR

HDFS Dataset

- Raw trip data: 17.25 GB
- Total files: 108 Parquet files
- Block size: 128 MB
- 100% healthy HDFS blocks

Data Flow

Raw Parquet / JSON → HDFS → Scala Spark
→ Snappy Parquet → HDFS

Results & Analysis

Full-Scale Multi-Source Ingestion & Veracity Preprocessing

Dataset Service	Raw Input Footprint	Clean Records / Coverage	Status / Veracity
Yellow Taxi (2022–2024)	1.80 GB (11.24M Rows)	10,580,210 Clean Rows	94.13% Retention
Green Taxi (2022–2024)	49.5 MB (898.9K Rows)	821,909 Clean Rows	91.43% Retention
FHVHV Uber/Lyft (2022–2024)	15.40 GB (36 Months)	36 Monthly Partitions	100% Ingested
Open-Meteo Weather JSON	2.75 MB (26.3K Hours)	Hourly Temp, Rain, Wind	Broadcast Joined
NYC Taxi Zone Lookup CSV	12.04 KB (265 Zones)	Borough, Zone, Service	Broadcast Joined
Total Multi-Source Corpus	17.25 GB Raw	11,402,119 Clean Unified Rows	100% Validated

Output Storage in HDFS

- 36 Monthly Partitions (2022–2024)
- Snappy Parquet: 318.2 MB
- Replicated Size: 954.7 MB (3×)

Validation

- Scala engine verified vs PySpark
- All 36 months verified active
- Zero missing keys / null timestamps

Challenges / Limitations

- **Schema Desync Across Years:** TLC changed physical column types in 2023 Parquet files (INT64 to INT32), requiring explicit casting in `SchemaUnifier.scala`.
- **Month-by-Month Ingestion:** Wildcard scans across multi-year folders caused RDD inference errors; solved by month-by-month Spark reader loop.
- **Driver Memory Management:** Processing 12M+ records required driver memory tuning (`--driver-memory 4g`) to prevent OutOfMemory errors.
- **Data Veracity Noise:** 737,003 records contained negative fares, zero distances, or taximeter glitches requiring 13 explicit cleaning rules.
- **Container Networking:** YARN containers required explicit RPC hostname resolution in Docker.

Big Data Tools Used

Tool	Purpose
Hadoop HDFS	Distributed storage of raw and processed trip datasets
Hadoop YARN	Cluster resource management and container scheduling
Apache Spark	Distributed data cleaning, SQL transformation, and analytics
Scala 2.12	High-performance compiled engine code
Apache Parquet	Columnar compressed storage for fast query access
Docker Compose	Multi-node cluster container deployment

Impact

- Processes 12.1+ million trip rows using distributed parallel tasks across 3 DataNodes.
- Snappy Parquet partitioning by year and month reduces query scan times by up to 50x.
- HDFS replication factor 3 provides high fault tolerance against node failures.

Impact Comparison: Big Data vs Conventional

Aspect	Conventional Single-Machine	Our TaxiPulse Platform
Data Size	Limited by local RAM (<8 GB)	Distributed storage across HDFS (100+ GB)
Processing	Single-process (Pandas/Excel)	Parallel Spark tasks across YARN nodes
Storage	Uncompressed raw CSV	Columnar Snappy Parquet
Data Access	Full-row linear scan	Column pruning & partition pruning
Partitioning	Full file scan	Year/month partition pruning
Fault Tolerance	None (Process crashes on error)	3x HDFS block replication
Scaling	Buy bigger single machine	Scale out by adding DataNodes

Overall Impact

Distributed HDFS storage + parallel Spark processing + columnar Parquet partitioning enables large-scale real-time urban mobility analytics.

References

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Thank You

Questions & Suggestions Welcome

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